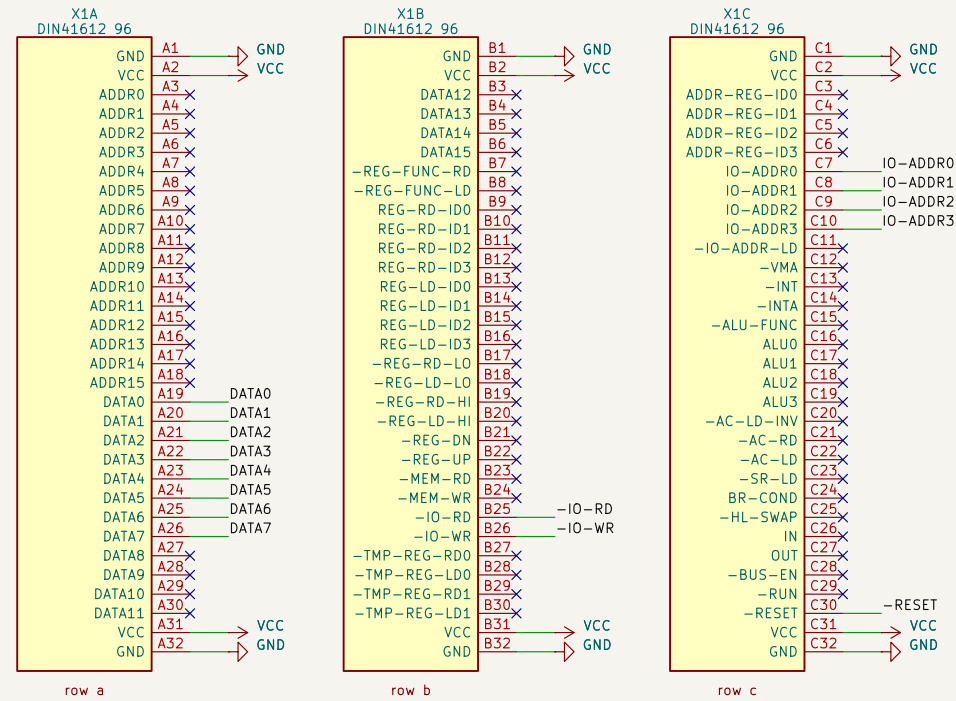
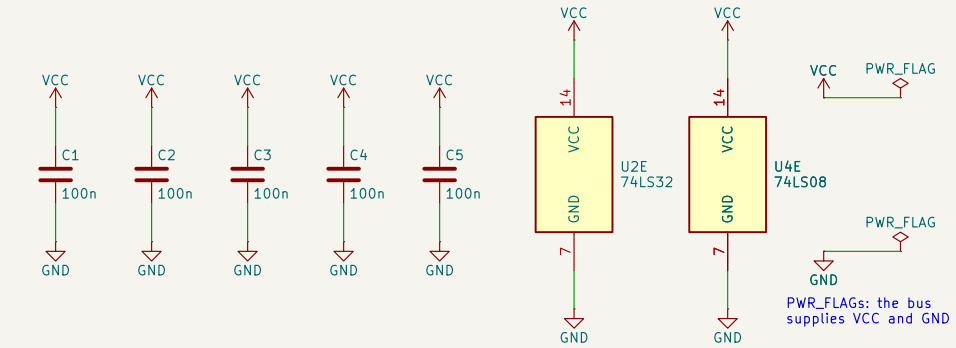


BUS X1 DIN 41612 (YACC1 V3.2)



Unused bus pins carry no-connect flags. The card uses DATA0-7, IO-ADDR0-3, -IO-RD, -IO-WR, -RESET and the six VCC / six GND pins.

POWER + DECOUPLING (one 100 nF per IC, at its VCC pin)

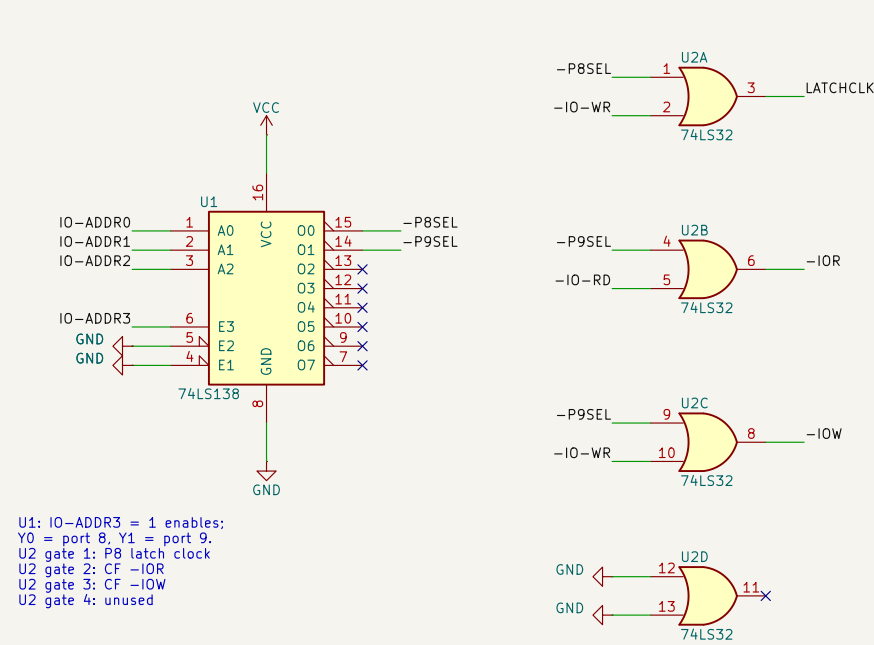


DESIGN NOTES

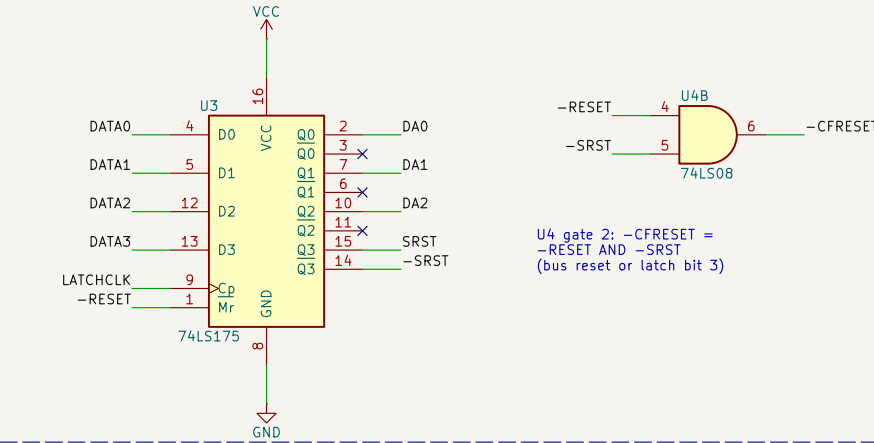
- Two I/O ports. P8 (write): 74LS175 latch U3 - bits 0-2 = the ATA task-file register (CF DA0-2), bit 3 = CF reset (1 = held in reset). P9 (read/write): the selected ATA register, 8-bit True IDE, through the 74LS245 U5.
- Decode: 74LS138 U1 on IO-ADDR0-2, enabled by IO-ADDR3 (Y0 = port 8, Y1 = port 9). 74LS32 U2 ORs each port select with -IO-WR / -IO-RD, so -IOR, -IOW and the latch clock are low only during the bus strobe for this card's port (the microcode sets IOADDR two steps before the strobe and holds it through the strobe).
- CS0 (J1 pin 37) is tied LOW and -CS1 (pin 38) HIGH: the strobes alone define every cycle. Gating -CS0 with the port-9 decode would release it on the same clock edge as -IO-RD at the end of an INP, a gate delay BEFORE -IOR rises, breaking the CF chip-select hold time (t9) on the data-register reads that advance its sector buffer.
- The CF card sits in a commercial CF-to-IDE adapter on the 40-pin header J1. The adapter is powered from J2 (floppy-style 4-pin: 1 = +5 V, 2 and 3 = GND, 4 = +12 V, not used) or, with JP1 fitted, from IDE pin 20 for adapters that take their power there.
- Pull-ups: CF D0-7 (RN1: an empty adapter reads \$FF, which the ROM driver treats as 'no card'), IORDY, -PDIAG, -DASP; -DMACK held inactive; CSEL grounded (master).
- LEDs: PWR; ACT = any port-9 access (the buffer enable); DASP = the CF's own activity/busy pin.
- SRST (U3 Q3, latch bit 3 true output) is a named net with no other connection (a probe point).

Theory of operation: docs/cards/cf.md. Circuit source: cf_netlist.py (edit it, then re-run build.sh).

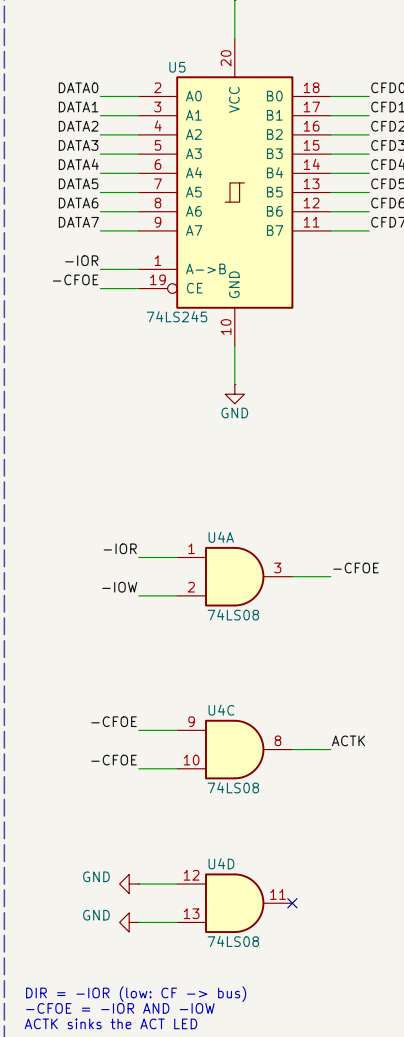
PORT DECODE + STROBE GATING



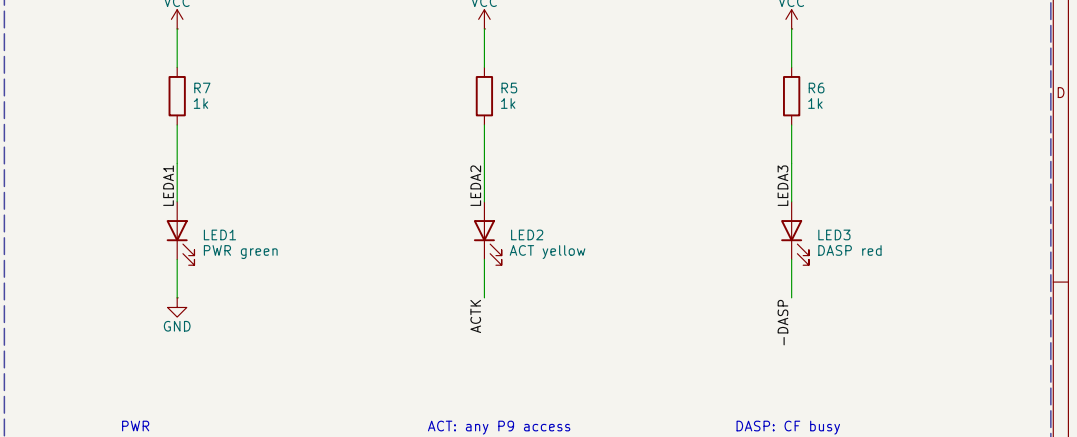
P8 LATCH: DA0-2 + CF RESET



DATA BUFFER



LEDS



IDE HEADER J1 (CF-TO-IDE ADAPTER)

